20260922 Quiz

for (MATH230) Probability & Statistics

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Exercise 3.1. Classify each random variable as discrete or continuous: X, the annual number of car accidents in Virginia; Y, the duration of 18 holes of golf; M, annual milk yield of one cow; N, eggs laid by a hen in a month; P, monthly building permits issued in a city; Q, weight of grain harvested per acre.

Solution. The counts X, N, P are $\boxed{\text{discrete}}$. The measurements Y, M, Q are $\boxed{\text{continuous}}$ (in the usual measurement models).

□

Exercise 3.3. A coin is tossed three times. Let W be the number of heads minus the number of tails. List the sample space and assign a value of W to every outcome.

Solution.

ω	HHH	HHT	HTH	THH	HTT	THT	TTH	TTT
$W(\omega)$	3	1	1	1	-1	-1	-1	-3

The eight outcomes displayed in the table constitute the sample space S.

□

Exercise 3.4. A fair coin is tossed until two consecutive heads appear. List every possible terminal outcome requiring at most six tosses. Is the complete sample space discrete? Explain.

Solution. The terminal strings of length at most six are

$$HH,\ THH,\ HTHH,\ TTHH,\ HTTHH,\ THTHH,\ TTTHH,$$
$$HTHTHH,\ HTTTHH,\ THTTHH,\ TTHTHH,\ TTTTHH.$$

Each string ends in HH with no earlier consecutive HH. The complete sample space is countably infinite and is therefore $\boxed{\text{discrete}}$.

□

Exercise 3.5. Find the normalization constant c for each discrete pmf on $x = 0, 1, 2$:

(a) $p_X(x) = c(x^2 + 4)$;

(b) $p_X(x) = c\binom{3}{x}\binom{3}{3-x}$.

Solution.

$$\text{(a)} \quad 1 = c(4 + 5 + 8), \qquad c = \boxed{\frac{1}{17}},$$

$$\text{(b)} \quad 1 = c(1 + 9 + 9), \qquad c = \boxed{\frac{1}{19}}.$$

□

Exercise 3.8. A coin is tossed three times independently, with a head twice as likely as a tail. Let W be the number of heads minus the number of tails. Find the probability distribution of W.

Solution. Here $\mathbb{P}(H) = 2/3$ and $\mathbb{P}(T) = 1/3$. The number of heads is binomial with parameters 3 and 2/3, and $W = 2H - 3$. Thus

w	-3	-1	1	3
$p_W(w)$	$1/27$	$6/27$	$12/27$	$8/27$

□

Exercise 3.9. A continuous response proportion X has density $f_X(x) = 2(x+2)/5$ for $0 < x < 1$, and zero otherwise.

(a) Verify that it is a density.

(b) Find $\mathbb{P}(1/4 < X < 3/4)$.

Solution.

$$\text{(a)} \quad \int_0^1 \frac{2(x+2)}{5}\,\mathrm{d}x = \frac{2}{5}\left[\frac{x^2}{2} + 2x\right]_0^1 = \boxed{1},$$

$$\text{(b)} \quad \int_{1/4}^{3/4} \frac{2(x+2)}{5}\,\mathrm{d}x = \boxed{\frac{1}{2}}.$$

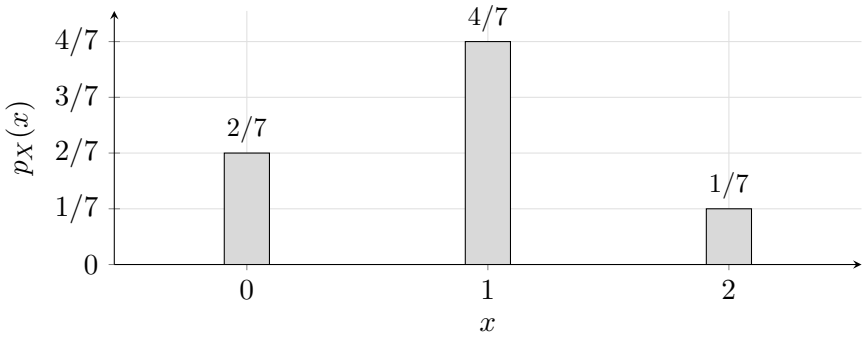
□

Exercise 3.11. Of seven television sets, two are defective. A hotel chooses three sets uniformly without replacement. Find the probability distribution of X, the number of defective sets purchased, and display its probability histogram.

Solution. The hypergeometric counting formula gives

$$p_X(x) = \frac{\binom{2}{x}\binom{5}{3-x}}{\binom{7}{3}}, \quad x = 0, 1, 2.$$

Hence $p_X(0) = 2/7$, $p_X(1) = 4/7$, and $p_X(2) = 1/7$. The corresponding probability histogram is



□

Exercise 3.12. The maturity T (years) of a bond has cumulative distribution function

$$F_T(t) = \begin{cases} 0, & t < 1, \\ 1/4, & 1 \le t < 3, \\ 1/2, & 3 \le t < 5, \\ 3/4, & 5 \le t < 7, \\ 1, & t \ge 7. \end{cases}$$

Find:

(a) $\mathbb{P}(T = 5)$.
(b) $\mathbb{P}(T > 3)$.
(c) $\mathbb{P}(1.4 < T < 6)$.
(d) $\mathbb{P}(T \le 5 \mid T \ge 2)$.

Solution. The jumps at $t = 1, 3, 5, 7$ are each $1/4$. Therefore

(a) $\boxed{1/4}$, (b) $\boxed{1/2}$,
(c) $\boxed{1/2}$, (d) $\dfrac{\mathbb{P}(T = 3 \text{ or } 5)}{\mathbb{P}(T = 3, 5, \text{ or } 7)} = \boxed{2/3}$.

□

Exercise 3.13. The number X of imperfections per 10 meters of synthetic fabric has pmf

x	0	1	2	3	4
$p_X(x)$	0.41	0.37	0.16	0.05	0.01

Construct its cdf.

Solution. Summing the probabilities up to each point gives

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 0.41, & 0 \le x < 1, \\ 0.78, & 1 \le x < 2, \\ 0.94, & 2 \le x < 3, \\ 0.99, & 3 \le x < 4, \\ 1, & x \ge 4. \end{cases}$$

□

Exercise 3.14. The waiting time X (hours) between speeders has cdf $F_X(x) = 0$ for $x < 0$ and $F_X(x) = 1 - e^{-4x}$ for $x \geq 0$. Find the probability of a wait of less than 12 minutes:

(a) from the cdf.

(b) from the pdf.

Solution. Since 12 minutes is 1/5 hour,

$$\text{(a)} \quad F_X(1/5) = \boxed{1 - e^{-4/5}},$$

$$\text{(b)} \quad \int_0^{1/5} 4e^{-4x}\, \mathrm{d}x = \boxed{1 - e^{-4/5}}.$$

Both values are about 0.5507.

□

Exercise 3.17. The continuous variable X has constant density $f_X(x) = 1/2$ for $1 < x < 3$, and zero otherwise.

(a) Verify normalization.

(b) Find $\mathbb{P}(2 < X < 2.5)$.

(c) Find $\mathbb{P}(X \leq 1.6)$.

Solution.

$$\text{(a)} \quad \int_1^3 \frac{1}{2}\, \mathrm{d}x = \boxed{1},$$

$$\text{(b)} \quad \int_2^{5/2} \frac{1}{2}\, \mathrm{d}x = \boxed{\frac{1}{4}},$$

$$\text{(c)} \quad \int_1^{8/5} \frac{1}{2}\, \mathrm{d}x = \boxed{\frac{3}{10}}.$$

□

Exercise 3.18. The density of X is $f_X(x) = 2(1 + x)/27$ for $2 < x < 5$, and zero otherwise. Find:

(a) $\mathbb{P}(X < 4)$.

(b) $\mathbb{P}(3 \leq X < 4)$.

Solution. An antiderivative is $(x^2 + 2x)/27$. Thus

$$\text{(a)} \quad \left[\frac{x^2 + 2x}{27}\right]_2^4 = \boxed{\frac{16}{27}},$$

$$\text{(b)} \quad \left[\frac{x^2 + 2x}{27}\right]_3^4 = \boxed{\frac{1}{3}}.$$

□

Exercise 3.21. Let $f_X(x) = k\sqrt{x}$ for $0 \leq x < 4$, and zero otherwise.

(a) Determine k.

(b) Obtain $F_X(x)$ and calculate $\mathbb{P}(3 < X < 4)$.

Solution. (a) Normalization gives

$$1 = k\int_0^4 \sqrt{x}\,\mathrm{d}x = \frac{16}{3}k,$$

so $\boxed{k = 3/16}$.

(b) The cdf is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x^{3/2}/8, & 0 \le x < 4, \\ 1, & x \ge 4. \end{cases}$$

Therefore $\mathbb{P}(3 < X < 4) = 1 - F_X(3) = \boxed{1 - 3\sqrt{3}/8}$.

□

Exercise 3.23. A biased coin has $\mathbb{P}(H) = 2/3$ and is tossed three times independently. Let W be heads minus tails. Determine its cdf and use it to calculate:

(a) $\mathbb{P}(W > 0)$.
(b) $\mathbb{P}(-1 \le W < 3)$.

Solution. As W takes $-3, -1, 1, 3$ with respective probabilities $1/27, 6/27, 12/27, 8/27$,

$$F_W(w) = \begin{cases} 0, & w < -3, \\ 1/27, & -3 \le w < -1, \\ 7/27, & -1 \le w < 1, \\ 19/27, & 1 \le w < 3, \\ 1, & w \ge 3. \end{cases}$$

Hence

(a) $\boxed{\mathbb{P}(W > 0) = \frac{20}{27}}$, (b) $\boxed{\mathbb{P}(-1 \le W < 3) = \frac{2}{3}}$.

□

Exercise 3.27. An electronic component has lifetime density $f_X(x) = e^{-x/1000}/1000$ for $x \ge 0$, and zero otherwise.

(a) Find $F_X(x)$.
(b) Find the probability of surviving beyond 1000 hours.
(c) Find the probability of failing before 2000 hours.

Solution. (a) For $x \ge 0$,

$$F_X(x) = \int_0^x \frac{e^{-t/1000}}{1000}\,\mathrm{d}t = 1 - e^{-x/1000},$$

and $F_X(x) = 0$ for $x < 0$.

(b) The probability of surviving beyond 1000 hours is $\boxed{e^{-1}}$.

(c) The probability of failing before 2000 hours is $\boxed{1 - e^{-2}}$.

□

Exercise 3.28. The weight X (ounces) of cereal in a box has density $f_X(x) = 2/5$ for $23.75 \leq x \leq 26.25$, and zero otherwise.

(a) Verify the density.
(b) Find $\mathbb{P}(X < 24)$.
(c) Find $\mathbb{P}(X > 26)$.

Solution.

$$\text{(a)} \quad \frac{2}{5}(26.25 - 23.75) = \boxed{1},$$
$$\text{(b)} \quad \frac{2}{5}(24 - 23.75) = \boxed{0.10},$$
$$\text{(c)} \quad \frac{2}{5}(26.25 - 26) = \boxed{0.10}.$$

□

Exercise 3.30. Measurements of physical systems are subject to variation, and modeling measurement error is important in statistics. Suppose that the error X in measuring a physical quantity has density $f_X(x) = k(3 - x^2)$ on $-1 \leq x \leq 1$, and zero otherwise.

(a) Determine k.
(b) Calculate $\mathbb{P}(X < 1/2)$.
(c) Calculate $\mathbb{P}(|X| > 0.8)$.

Solution.

$$1 = k \int_{-1}^{1} (3 - x^2)\, \mathrm{d}x = \frac{16k}{3}, \qquad \boxed{k = \frac{3}{16}}.$$

Consequently,

$$\text{(b)} \quad \frac{3}{16} \int_{-1}^{1/2} (3 - x^2)\, \mathrm{d}x = \boxed{\frac{99}{128}},$$
$$\text{(c)} \quad 2 \cdot \frac{3}{16} \int_{4/5}^{1} (3 - x^2)\, \mathrm{d}x = \boxed{\frac{41}{250}}.$$

□

Exercise 3.32. A company's proportion Y of its budget devoted to pollution control has density $f_Y(y) = 5(1 - y)^4$ on $0 \leq y \leq 1$, and zero elsewhere. Verify that this is a density and find $\mathbb{P}(Y < 0.10)$ and $\mathbb{P}(Y > 0.50)$.

Solution.

$$\text{(a)} \quad \int_0^1 5(1 - y)^4\, \mathrm{d}y = \boxed{1},$$
$$\text{(b)} \quad 1 - (0.90)^5 = \boxed{0.40951},$$
$$\text{(c)} \quad (0.50)^5 = \boxed{\frac{1}{32}}.$$

□

Exercise 3.36. The variable X has density $f_X(x) = 2(1-x)$ on $0 < x < 1$, and zero elsewhere. Find:

(a) $\mathbb{P}(X < 0.5)$.
(b) $\mathbb{P}(X > 0.4)$.
(c) $\mathbb{P}(X < 0.7 \mid X \geq 0.5)$.

Solution. Since $\mathbb{P}(X \geq t) = (1-t)^2$ for $0 \leq t \leq 1$,

$$\begin{aligned} &\text{(a)} \quad 1 - (0.5)^2 = \boxed{0.75}, \\ &\text{(b)} \quad (0.6)^2 = \boxed{0.36}, \\ &\text{(c)} \quad \frac{(0.5)^2 - (0.3)^2}{(0.5)^2} = \boxed{0.64}. \end{aligned}$$

□

Exercise 3.37. Find the constants c for which the following joint mass functions are normalized:

(a) $p_{X,Y}(x,y) = cxy$, for $x, y \in \{1, 2\}$;
(b) $p_{X,Y}(x,y) = c|x-y|$, for $x \in \{-2, 0, 2\}$, $y \in \{-3, 3\}$.

Solution.

$$\begin{aligned} &\text{(a)} \quad 1 = c\left(\sum_{x=1}^{2} x\right)^2 = 9c, && \boxed{c = 1/9}, \\ &\text{(b)} \quad 1 = c(1+5+3+3+5+1) = 18c, && \boxed{c = 1/18}. \end{aligned}$$

□

Exercise 3.38. The joint mass function is $p_{X,Y}(x,y) = (x+y)/30$ for $x = 0, 1, 2, 3$ and $y = 0, 1, 2$ (zero elsewhere). Find

(a) $\mathbb{P}(X \leq 1, Y = 1)$;
(b) $\mathbb{P}(X > 1, Y \leq 1)$;
(c) $\mathbb{P}(X \leq Y)$;
(d) $\mathbb{P}(X + Y = 2)$.

Solution. Summing the indicated joint masses,

$$\begin{aligned} &\text{(a)} \quad \frac{(0+1)+(1+1)}{30} = \boxed{\frac{1}{10}}, \\ &\text{(b)} \quad \frac{(2+0)+(2+1)+(3+0)+(3+1)}{30} = \boxed{\frac{2}{5}}, \\ &\text{(c)} \quad \frac{(0+1)+(1+1)+(0+2)+(1+2)+(2+2)}{30} = \boxed{\frac{2}{5}}, \\ &\text{(d)} \quad \frac{(0+2)+(1+1)+(2+0)}{30} = \boxed{\frac{1}{5}}. \end{aligned}$$

□

Exercise 3.39. A random sample of four fruits is drawn without replacement from three oranges, two apples, and three bananas. Let X and Y be the numbers of oranges and apples selected. Find:

(a) the joint pmf explicitly.
(b) $\mathbb{P}(X + Y \le 2)$.

Solution. (a) The joint mass is

$$\boxed{p_{X,Y}(x,y) = \frac{\binom{3}{x}\binom{2}{y}\binom{3}{4-x-y}}{\binom{8}{4}}}$$

for integers $0 \le x \le 3$, $0 \le y \le 2$, $1 \le x + y \le 4$, and $p_{X,Y}(x,y) = 0$ otherwise. (b) The event $X + Y \le 2$ means at least two bananas are drawn, so

$$\mathbb{P}(X + Y \le 2) = \frac{\binom{3}{2}\binom{5}{2} + \binom{3}{3}\binom{5}{1}}{\binom{8}{4}} = \frac{35}{70} = \boxed{\frac{1}{2}}.$$

□

Exercise 3.40. The proportions of time a fast-food restaurant's drive-through and walk-in services are busy are X and Y. Their joint density is $f_{X,Y}(x,y) = \frac{2}{3}(x + 2y)$ on $0 < x < 1$, $0 < y < 1$, and zero otherwise. Find:

(a) f_X.
(b) f_Y.
(c) $\mathbb{P}(X < 1/2)$.

Solution. (a) Integrating over y gives

$$\boxed{f_X(x) = \frac{2}{3}(x+1)}, \qquad 0 < x < 1.$$

(b) Integrating over x gives

$$\boxed{f_Y(y) = \frac{1}{3}(1+4y)}, \qquad 0 < y < 1.$$

Both densities are zero outside $(0,1)$.

(c) Therefore

$$\mathbb{P}(X < 1/2) = \int_0^{1/2} \frac{2}{3}(x+1)\,\mathrm{d}x = \boxed{\frac{5}{12}}.$$

□

Exercise 3.41. In a one-kilogram box of chocolates, X and Y are the weights in kilograms of creams and toffees; the remainder is cordials. Assume the joint density $f_{X,Y}(x,y) = 24xy$ for $x, y \ge 0$, $x + y \le 1$, and zero elsewhere. Find

(a) the probability that cordials exceed half the box weight;
(b) the marginal density of X;
(c) $\mathbb{P}(Y < 1/8 \mid X = 3/4)$.

Solution. (a) Cordials exceed 1/2 precisely when $X + Y < 1/2$. Since the integrand is homogeneous of degree two, scaling the triangle by 1/2 multiplies the integral by $(1/2)^4$; hence $\boxed{1/16}$.

(b)

$$f_X(x) = \int_0^{1-x} 24xy \, \mathrm{d}y = \boxed{12x(1-x)^2}, \quad 0 < x < 1.$$

(c) With $X = 3/4$, the conditional support of Y is $(0, 1/4)$ and its conditional density is proportional to y. Therefore

$$\mathbb{P}(Y < 1/8 \mid X = 3/4) = \frac{(1/8)^2}{(1/4)^2} = \boxed{\frac{1}{4}}.$$

□

Exercise 3.42. The lifetimes X and Y (years) of two components have joint density $f_{X,Y}(x, y) = e^{-(x+y)}$ for $x, y > 0$, and zero otherwise. Find $\mathbb{P}(1 < X < 2 \mid Y = 2)$.

Solution. The joint density factorizes into two exponential marginals, so X and Y are independent. Therefore

$$\mathbb{P}(1 < X < 2 \mid Y = 2) = \int_1^2 e^{-x} \, \mathrm{d}x = \boxed{e^{-1} - e^{-2}}.$$

□

Exercise 3.43. Let X be a reaction time (seconds) and Y a temperature. Their joint density is $f_{X,Y}(x, y) = 4xy$ on $0 < x < 1$, $0 < y < 1$ and zero elsewhere. Find:

(a) $\mathbb{P}(0 \le X \le 1/2,\ 1/4 \le Y \le 1/2)$.
(b) $\mathbb{P}(X < Y)$.

Solution. The density factorizes as $(2x)(2y)$, and the marginals have the same form. Thus

$$\text{(a)} \quad \left(\int_0^{1/2} 2x \, \mathrm{d}x \right) \left(\int_{1/4}^{1/2} 2y \, \mathrm{d}y \right) = \boxed{\frac{3}{64}},$$

$$\text{(b)} \quad \int_0^1 \int_0^y 4xy \, \mathrm{d}x \, \mathrm{d}y = \boxed{\frac{1}{2}}.$$

□

Exercise 3.46. The joint pmf $p_{X,Y}(x, y) = (x + y)/30$ is supported on $x \in \{0, 1, 2, 3\}$ and $y \in \{0, 1, 2\}$. Find the marginal pmfs of X and Y.

Solution.

$$p_X(x) = \sum_{y=0}^{2} \frac{x+y}{30} = \boxed{\frac{x+1}{10}}, \quad x = 0, 1, 2, 3,$$

$$p_Y(y) = \sum_{x=0}^{3} \frac{x+y}{30} = \boxed{\frac{3+2y}{15}}, \quad y = 0, 1, 2.$$

Both functions are zero outside their displayed supports.

□

Exercise 3.47. For the daily amount Y of kerosene available and the amount X sold (both in thousands of liters), the joint density is $f_{X,Y}(x,y) = 2$ for $0 < x \le y < 1$, and zero otherwise.

(a) Are X, Y independent?
(b) Find $\mathbb{P}(1/4 < X < 1/2 \mid Y = 3/4)$.

Solution. (a) The triangular support rules out independence. Indeed, $f_X(x) = 2(1-x)$ and $f_Y(y) = 2y$, while $f_{X,Y}(x,y)$ is not their product. Thus $\boxed{\text{dependent}}$.

(b) Since $f_{X|Y}(x \mid y) = 1/y$ for $0 < x < y$,

$$\mathbb{P}(1/4 < X < 1/2 \mid Y = 3/4) = \int_{1/4}^{1/2} \frac{4}{3}\, \mathrm{d}x = \boxed{\frac{1}{3}}.$$

□

Exercise 3.48. A random sample of four fruits is drawn without replacement from three oranges, two apples, and three bananas. Let X and Y be the numbers of oranges and apples selected. The joint pmf is

$$p_{X,Y}(x,y) = \frac{\binom{3}{x}\binom{2}{y}\binom{3}{4-x-y}}{\binom{8}{4}}$$

for admissible integer pairs (x,y), and zero otherwise. Find:

(a) the conditional pmf $p_{Y|X}(y \mid 2)$.
(b) $\mathbb{P}(Y = 0 \mid X = 2)$.

Solution. (a) For $X = 2$, the unnormalized counts for $Y = 0, 1, 2$ are $9, 18, 3$, totaling 30. Thus

$$\boxed{p_{Y|X}(y \mid 2) = \begin{cases} 3/10, & y = 0, \\ 3/5, & y = 1, \\ 1/10, & y = 2, \\ 0, & \text{otherwise}, \end{cases}}$$

(b) Hence $\boxed{\mathbb{P}(Y = 0 \mid X = 2) = 3/10}$.

□

Exercise 3.49. The joint pmf of the number X of machine malfunctions and the number Y of technician calls is

$p_{X,Y}(x,y)$	$x = 1$	$x = 2$	$x = 3$
$y = 1$	0.05	0.05	0.10
$y = 3$	0.05	0.10	0.35
$y = 5$	0.00	0.20	0.10

Find:

(a) the marginal pmf of X.
(b) that of Y.
(c) $\mathbb{P}(Y = 3 \mid X = 2)$.

Solution. (a) Summing over y gives the marginal of X:

x	1	2	3
$p_X(x)$	0.10	0.35	0.55

(b) Summing over x gives the marginal of Y:

y	1	3	5
$p_Y(y)$	0.20	0.50	0.30

(c) Therefore $\mathbb{P}(Y = 3 \mid X = 2) = 0.10/0.35 = \boxed{2/7}$.

□

Exercise 3.50. The joint pmf of X, Y is

$p_{X,Y}(x, y)$	$x = 2$	$x = 4$
$y = 1$	0.10	0.15
$y = 3$	0.20	0.30
$y = 5$	0.10	0.15

Find the marginal pmfs of X and Y.

Solution. The marginal distributions are

$$p_X(2) = 0.40, \qquad p_X(4) = 0.60,$$
$$p_Y(1) = 0.25, \quad p_Y(3) = 0.50, \quad p_Y(5) = 0.25.$$

Both pmfs are zero elsewhere.

□

Exercise 3.52. A coin is tossed twice independently with $\mathbb{P}(H) = 0.30$. Let Z be the indicator of heads on the first toss, and W the total number of heads. Find:

(a) the joint pmf of (W, Z).
(b) the marginal pmf of W.
(c) the marginal pmf of Z.
(d) the probability of at least one head.

Solution. (a) The outcomes HH, HT, TH, TT have probabilities $0.09, 0.21, 0.21, 0.49$, respectively. Therefore

$p_{W,Z}(w, z)$	$z = 0$	$z = 1$
$w = 0$	0.49	0
$w = 1$	0.21	0.21
$w = 2$	0	0.09

(b) The marginal pmf of W is $p_W(0) = 0.49$, $p_W(1) = 0.42$, $p_W(2) = 0.09$.

(c) The marginal pmf of Z is $p_Z(0) = 0.70$, $p_Z(1) = 0.30$.

(d) Finally,

$$\mathbb{P}(W \geq 1) = 1 - \mathbb{P}(W = 0) = \boxed{0.51}.$$

□

Exercise 3.53. The joint density of X and Y is $f_{X,Y}(x,y) = (6-x-y)/8$ for $0 < x < 2$ and $2 < y < 4$, and zero otherwise. Find $\mathbb{P}(1 < Y < 3 \mid X = 1.5)$.

Solution. Since

$$f_X(x) = \int_2^4 \frac{6-x-y}{8}\, \mathrm{d}y = \frac{3-x}{4},$$

the conditional density at $x = 3/2$ is

$$f_{Y|X}(y \mid 3/2) = \frac{(9/2-y)/8}{(3-3/2)/4} = \frac{9/2-y}{3}, \quad 2 < y < 4.$$

Thus

$$\mathbb{P}(1 < Y < 3 \mid X = 3/2) = \int_2^3 \frac{9/2-y}{3}\, \mathrm{d}y = \boxed{\frac{2}{3}}.$$

□

Exercise 3.54. For the joint pmf

$p_{X,Y}(x,y)$	$x=1$	$x=2$	$x=3$
$y=1$	0.05	0.05	0.10
$y=3$	0.05	0.10	0.35
$y=5$	0.00	0.20	0.10

Determine whether X and Y are independent.

Solution. The marginals satisfy $p_X(1) = 0.10$ and $p_Y(1) = 0.20$, but

$$p_{X,Y}(1,1) = 0.05 \neq (0.10)(0.20) = 0.02.$$

Therefore $\boxed{X \text{ and } Y \text{ are dependent}}$.

□

Exercise 3.55. For the joint pmf

$p_{X,Y}(x,y)$	$x=2$	$x=4$
$y=1$	0.10	0.15
$y=3$	0.20	0.30
$y=5$	0.10	0.15

Determine whether X and Y are independent.

Solution. The marginal probabilities are

$$p_X(2) = 0.40, \quad p_X(4) = 0.60, \qquad (p_Y(1), p_Y(3), p_Y(5)) = (0.25, 0.50, 0.25).$$

Every joint mass equals the product of the corresponding marginals. Hence

$$\boxed{X \text{ and } Y \text{ are independent.}}$$

□

Exercise 3.56. The joint density of X, Y is $f_{X,Y}(x,y) = 6x$ for $0 < x < 1$, $0 < y < 1 - x$, and zero otherwise.

(a) Show dependence.
(b) Find $\mathbb{P}(X > 0.3 \mid Y = 0.5)$.

Solution. (a) The joint density has triangular support, unlike the product of two positive marginals on a rectangle. Therefore the variables are $\boxed{\text{dependent}}$.

(b) The conditional density for $Y = 1/2$ is proportional to x on $0 < x < 1/2$. Thus

$$\mathbb{P}(X > 0.3 \mid Y = 0.5) = \frac{\int_{0.3}^{0.5} x \, \mathrm{d}x}{\int_{0}^{0.5} x \, \mathrm{d}x} = \boxed{0.64}.$$

□

Exercise 3.57. The joint density is $f_{X,Y,Z}(x,y,z) = kxy^2z$ on $0 < x < 1$, $0 < y < 1$, $0 < z < 2$, and zero otherwise.

(a) Determine k.
(b) Find $\mathbb{P}(X < 1/4, Y > 1/2, 1 < Z < 2)$.

Solution. (a) Normalization gives

$$1 = k\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)(2) = \frac{k}{3}, \quad \boxed{k = 3}.$$

(b) The density factors, so

$$\mathbb{P}(X < 1/4, Y > 1/2, 1 < Z < 2) = \left(\frac{1}{16}\right)\left(\frac{7}{8}\right)\left(\frac{3}{4}\right) = \boxed{\frac{21}{512}}.$$

□

Exercise 3.60. The joint density of X, Y, Z is $f_{X,Y,Z}(x,y,z) = \frac{4}{9}xyz^2$ for $0 < x, y < 1$ and $0 < z < 3$, and zero otherwise. Find

(a) the joint marginal density of (X, Y);
(b) the marginal density of Z;
(c) $\mathbb{P}(1/4 < X < 1/2, Y > 1/3, 2 < Z < 3)$;
(d) $\mathbb{P}(0 < Z < 2 \mid X = 1/2, Y = 1/2)$.

Solution.

$$\text{(a)} \quad f_{X,Y}(x,y) = \int_0^3 \frac{4}{9}xyz^2 \, \mathrm{d}z = \boxed{4xy},$$

$$\text{(b)} \quad f_Z(z) = \int_0^1 \int_0^1 \frac{4}{9}xyz^2 \, \mathrm{d}x \, \mathrm{d}y = \boxed{\frac{z^2}{9}}.$$

These apply on $0 < x, y < 1$ and $0 < z < 3$ respectively, and vanish elsewhere. The density factors, giving

$$\text{(c)} \quad \left(\frac{3}{16}\right)\left(\frac{8}{9}\right)\left(\frac{19}{27}\right) = \boxed{\frac{19}{162}},$$

$$\text{(d)} \quad \int_0^2 \frac{z^2}{9} \, \mathrm{d}z = \boxed{\frac{8}{27}}.$$

□

Exercise 3.61. A one-unit tobacco blend contains Turkish and domestic tobacco in proportions X, Y, with the remainder being other tobacco. Their joint density is $f_{X,Y}(x,y) = 24xy$ for $x, y \geq 0$, $x + y \leq 1$, and zero elsewhere. Find:

(a) $\mathbb{P}(X > 1/2)$.

(b) the marginal density of Y.

(c) $\mathbb{P}(X < 1/8 \mid Y = 3/4)$.

Solution. (a) The marginal is $f_X(x) = 12x(1-x)^2$, so

$$\mathbb{P}(X > 1/2) = \int_{1/2}^{1} 12x(1-x)^2 \, \mathrm{d}x = \boxed{\frac{5}{16}}.$$

(b) Integrating over $0 < x < 1 - y$,

$$f_Y(y) = \int_0^{1-y} 24xy \, \mathrm{d}x = \boxed{12y(1-y)^2}, \quad 0 < y < 1.$$

(c) With $Y = 3/4$, the conditional density of X is proportional to x on $(0, 1/4)$, so

$$\mathbb{P}(X < 1/8 \mid Y = 3/4) = \left(\frac{1/8}{1/4}\right)^2 = \boxed{\frac{1}{4}}.$$

□

Exercise 3.68. The joint density is $f_{X,Y}(x,y) = (3x - y)/11$ on $1 < x < 3$ and $0 < y < 1$, and zero elsewhere.

(a) Find the marginal densities.

(b) Check independence.

(c) Find $\mathbb{P}(X > 2)$.

Solution. (a) Integrating gives

$$f_X(x) = \boxed{\frac{3x - 1/2}{11}}, \quad 1 < x < 3; \qquad f_Y(y) = \boxed{\frac{12 - 2y}{11}}, \quad 0 < y < 1.$$

Both vanish outside their stated supports.

(b) The density $(3x - y)/11$ does not factor into these marginals, so $\boxed{\text{dependent}}$.

(c)

$$\mathbb{P}(X > 2) = \int_2^3 \frac{3x - 1/2}{11} \, \mathrm{d}x = \boxed{\frac{7}{11}}.$$

□

Exercise 3.75. A chemical blend has component proportions X_1 and X_2 with joint density $f(x_1, x_2) = 2$ for $0 < x_1 < x_2 < 1$, and zero otherwise.

(a) Find f_{X_1}.
(b) Find f_{X_2}.
(c) Calculate $\mathbb{P}(X_1 < 0.2, X_2 > 0.5)$.
(d) Determine $f_{X_1|X_2}(x_1 \mid x_2)$.

Solution.

$$\text{(a)} \quad f_{X_1}(x_1) = \int_{x_1}^{1} 2 \, \mathrm{d}x_2 = \boxed{2(1 - x_1)}, \quad 0 < x_1 < 1,$$

$$\text{(b)} \quad f_{X_2}(x_2) = \int_{0}^{x_2} 2 \, \mathrm{d}x_1 = \boxed{2x_2}, \quad 0 < x_2 < 1,$$

$$\text{(c)} \quad \int_{1/2}^{1} \int_{0}^{1/5} 2 \, \mathrm{d}x_1 \, \mathrm{d}x_2 = \boxed{\frac{1}{5}},$$

$$\text{(d)} \quad f_{X_1|X_2}(x_1 \mid x_2) = \frac{2}{2x_2} = \boxed{\frac{1}{x_2}}, \quad 0 < x_1 < x_2 < 1.$$

□